

Name: Shashikala Rajesh Gupta	Roll Number: 12
Year/Dept: BE COMP 2026-27	Subject: Big Data Analytics (Assignment)

BDA assignment (Databricks)

The following five labs needs to be performed with screenshots for every lab

1. Spark SQL Basics

Import a CSV dataset (e.g., sales or student records) into Databricks Community Edition and run simple SQL queries: `SELECT, GROUP BY, ORDER BY`.

- *Learning outcome:* Transition from Hive SQL to Spark SQL in Databricks.

2. Delta Lake Time Travel

Create a small Delta table, update a few rows, and query older versions using `VERSION AS OF`.

- *Learning outcome:* Understand ACID transactions and schema evolution compared to Hadoop HDFS.

3. Streaming Data Simulation

Use a JSON file stream (simulated IoT sensor data) and process it with Spark Structured Streaming.

- *Learning outcome:* Compare with Flume/Kafka ingestion, but in a modern streaming pipeline.

4. DataFrame Transformations

Load a dataset into a Spark DataFrame, perform filtering, joins, and aggregations, then visualize results with built-in charts.

- *Learning outcome:* Reinforce Spark programming concepts beyond Sqoop/Hive.

5. Genie Agent Demo

Register a small dataset in Unity Catalog (e.g., sales or movie ratings), configure a Genie Agent, and ask natural-language queries like “Show top 5 products by revenue.”

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- *Learning outcome:* Experience conversational BI and see how SQL is auto-generated.

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Theory:

A Lakehouse Architecture is a modern hybrid data platform that combines the low-cost, scalable storage of a data lake with the reliability, governance, and performance of a data warehouse. It allows organizations to store raw structured, semi-structured, and unstructured data while still supporting ACID transactions, schema enforcement, and advanced analytics.

☐ Key Characteristics of Lakehouse Architecture

- **Unified Storage:** Uses cloud object storage (e.g., AWS S3, Azure Blob, GCP Storage) for raw data.
- **Open Table Format:** Layers like Delta Lake or Apache Iceberg add metadata, schema enforcement, and versioning.
- **ACID Transactions:** Ensures reliable reads/writes, unlike traditional data lakes.
- **Schema Evolution:** Pipelines adapt without breaking when data changes.
- **Governance:** Unified security and cataloging across structured and unstructured data.
- **Performance:** Warehouse-grade query speed with engines like Photon or Presto.

How It Works

1. **Storage Layer:** Raw data stored in low-cost cloud object storage.
2. **Metadata Layer:** Tracks schema, versions, and transactions (Delta Lake/Iceberg).
3. **Compute Layer:** Engines like Spark, Databricks SQL, or BI tools query the data.
4. **Medallion Architecture:** Organizes data into Bronze (raw), Silver (cleaned), and Gold (aggregated) layers for progressive quality.

Problems It Solves

- **Eliminates Silos:** No need to maintain separate data lakes and warehouses.
- **Reduces Duplication:** Same dataset serves raw exploration and BI analytics.
- **Improves Freshness:** Data is available for real-time streaming and batch pipelines.
- **Supports AI/ML:** Unified platform for BI, predictive analytics, and generative AI.

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Comparison: Data Lake vs Warehouse vs Lakehouse

Feature	Data Lake	Data Warehouse	Lakehouse
Data Types	Raw structured, semi-structured, unstructured	Structured only	All types unified
Cost	Low	High	Low (storage) + optimized compute
Transactions	No ACID	ACID supported	ACID supported
Governance	Weak	Strong	Unified
Use Cases	Raw storage, ML training	BI, reporting	BI + ML + streaming

✓ Why It Matters for You (Teaching Context)

Since your students already know Hadoop, Flume, and Sqoop:

- Data Lakehouse (Databricks) shows them the evolution beyond Hadoop.
- They'll see how Delta Lake improves over HDFS with ACID and time travel.
- They'll understand how Genie Agents and BI tools query the same unified dataset.
- This prepares them for industry demand, where Databricks and Lakehouse skills are highly valued.

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STEPS:

Lab 1: Spark SQL Basics in Databricks

1. Setup Databricks Community Edition

- Sign up at Databricks Community Edition (community.cloud.databricks.com in Bing).
- Create a new **Notebook** (Python or SQL).

2. Download Dataset

Use a simple public CSV dataset. For example:

- **Student Performance Dataset:** student_scores.csv (raw.githubusercontent.com in Bing)
- **Retail Sales Dataset:** retail_sales.csv (raw.githubusercontent.com in Bing)

☐ These are small, free datasets suitable for classroom labs.

3. Upload Dataset to Databricks

- In the left sidebar, go to **Data** → **Add Data** → **Upload File**.
- Upload `student_scores.csv` or `retail_sales.csv`.
- Databricks will store it in `/FileStore/tables/`.

4. Load Dataset into Spark DataFrame

python

Load CSV into DataFrame

```
df = spark.read.csv("/FileStore/tables/student_scores.csv", header=True,
inferSchema=True)
```

Show first 5 rows

```
df.show(5)
```

5. Create Temporary SQL View

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python

```
# Register DataFrame as SQL view
df.createOrReplaceTempView("students")
```

6. Run Spark SQL Queries

sql

```
-- Select all records
```

```
SELECT * FROM students;
```

```
-- Count students by gender
```

```
SELECT gender, COUNT(*) AS total_students
FROM students
GROUP BY gender;
```

```
-- Average math score by parental education
```

```
SELECT parental_level_of_education, AVG(math_score) AS avg_math
FROM students
GROUP BY parental_level_of_education
ORDER BY avg_math DESC;
```

```
-- Top 5 students by reading score
```

```
SELECT student_id, reading_score
FROM students
ORDER BY reading_score DESC
LIMIT 5;
```

7. Visualize Results

- After running a SQL cell, click **+** → **Visualization**.
- Choose **Bar chart** or **Pie chart** to visualize grouped results.
- Example: Average math score by parental education → Bar chart.

☐ **Learning Outcomes**

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- Students see how **Spark SQL** queries look similar to Hive SQL but run on Databricks.
- They learn to **import CSVs, create views, run queries, and visualize results.**
- This lab bridges their Hadoop background with modern **lakehouse workflows.**

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Lab 2: Delta Lake Time Travel

1. Setup Notebook

- Open Databricks Community Edition.
- Create a new notebook (Python).

2. Create a Sample Dataset

python

Create sample data

```
data = [  
    (1, "Alice", 85),  
    (2, "Bob", 90),  
    (3, "Charlie", 78)  
]
```

```
columns = ["student_id", "name", "score"]
```

```
df = spark.createDataFrame(data, columns)  
df.show()
```

3. Write Data to Delta Table

python

Save as Delta table

```
df.write.format("delta").mode("overwrite").save("/tmp/delta/students")
```

Read back

```
delta_df = spark.read.format("delta").load("/tmp/delta/students")  
delta_df.show()
```

4. Update Data (Versioning)

python

```
from delta.tables import DeltaTable
```

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```
# Load Delta table
delta_table = DeltaTable.forPath(spark, "/tmp/delta/students")
```

```
# Update score for Alice
delta_table.update(
    condition = "name = 'Alice'",
    set = { "score": "95" }
)
```

```
# Show updated data
delta_table.toDF().show()
```

5. Add New Record

```
python
new_data = [(4, "David", 88)]
new_df = spark.createDataFrame(new_data, columns)

new_df.write.format("delta").mode("append").save("/tmp/delta/students")

# Show current table
spark.read.format("delta").load("/tmp/delta/students").show()
```

6. Query Older Versions (Time Travel)

```
python
# Check history
delta_table.history().show()

# Read version 0 (original data)
spark.read.format("delta").option("versionAsOf", 0).load("/tmp/delta/students").show()

# Read version 1 (after Alice update)
spark.read.format("delta").option("versionAsOf", 1).load("/tmp/delta/students").show()
```

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7. Learning Outcomes

- Students see how **Delta Lake** supports ACID transactions.
- They learn **time travel**: querying older versions of data.
- They understand why Delta Lake is superior to plain HDFS (which lacks versioning and schema enforcement).

This lab clearly shows the **evolution from Hadoop HDFS → Delta Lake**.

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Lab 3: Streaming Data Simulation in Databricks

1. Setup Notebook

- Open Databricks Community Edition.
- Create a new notebook (Python).

2. Create a Simulated Streaming Source

We'll simulate streaming data by writing JSON files into a directory.

```
python
import time
import json

# Create a directory for streaming data
dbutils.fs.mkdirs("/tmp/streaming_data")

# Simulate writing JSON events
for i in range(5):
    event = {"event_id": i, "sensor_value": i*10, "timestamp": str(time.time())}
    path = f"/tmp/streaming_data/event_{i}.json"
    dbutils.fs.put(path, json.dumps(event), overwrite=True)
```

3. Read Stream with Structured Streaming

```
python
# Read streaming data
stream_df = (
    spark.readStream
    .format("json")
    .schema("event_id INT, sensor_value INT, timestamp STRING")
    .load("/tmp/streaming_data")
)

# Display stream in real-time
display(stream_df)
```

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4. Process Streaming Data

python

Aggregate sensor values

```
agg_df = (
    stream_df.groupBy("event_id")
    .agg({"sensor_value": "avg"})
    .withColumnRenamed("avg(sensor_value)", "avg_value")
)
```

Write results to memory table

```
query = (
    agg_df.writeStream
    .outputMode("complete")
    .format("memory")
    .queryName("sensor_avg")
    .start()
)
```

5. Query Results

sql

-- Run this in a SQL cell

```
SELECT * FROM sensor_avg;
```

6. Visualize Results

- Use **bar chart visualization** to show average sensor values by event_id.
- Students can see results update as new JSON files are added.

7. Learning Outcomes

- Students understand how **Structured Streaming** replaces Flume/Kafka ingestion pipelines.

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- They see **real-time aggregation** in Databricks.
- They learn how streaming data can be queried continuously, unlike batch Hadoop jobs.

This lab demonstrates the **shift from batch Hadoop ingestion → real-time streaming in Databricks**.

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Lab 4: DataFrame Transformations in Databricks

1. Setup Notebook

- Open Databricks Community Edition.
- Create a new notebook (Python).

2. Load Sample Dataset

We'll use a small CSV dataset (students or retail sales). Upload it via **Data** → **Add Data** → **Upload File**.

Example: `student_scores.csv` with columns: `student_id`, `name`, `gender`, `math_score`, `reading_score`, `writing_score`.

python

`# Load CSV into DataFrame`

```
df = spark.read.csv("/FileStore/tables/student_scores.csv", header=True,
inferSchema=True)
df.show(5)
```

3. Filter Rows

python

```
# Filter students with math_score > 80
high_math = df.filter(df.math_score > 80)
high_math.show()
```

4. Select Specific Columns

python

```
# Select name and reading_score
df.select("name", "reading_score").show(5)
```

5. Group and Aggregate

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```
python
# Average scores by gender
avg_scores = df.groupBy("gender").agg(
    {"math_score": "avg", "reading_score": "avg", "writing_score": "avg"}
)
avg_scores.show()
```

6. Join Two DataFrames

```
python
# Create another DataFrame with student hobbies
hobbies = [
    (1, "Football"),
    (2, "Chess"),
    (3, "Music"),
    (4, "Dance")
]
hobby_df = spark.createDataFrame(hobbies, ["student_id", "hobby"])

# Join with main DataFrame
joined_df = df.join(hobby_df, "student_id", "left")
joined_df.show()
```

7. Sort Results

```
python
# Top 5 students by writing score
df.orderBy(df.writing_score.desc()).show(5)
```

8. Visualize Results

- After running aggregations, click + → **Visualization**.
- Example: Bar chart of average math score by gender.

☐ **Learning Outcomes**

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- Students practice **filtering, selecting, grouping, joining, and sorting** with Spark DataFrames.
- They see how transformations are more flexible than Hive queries.
- They learn how to combine multiple datasets seamlessly.

This lab shows the **power of Spark DataFrames compared to Hadoop's batch-oriented tools.**

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Lab 5: Genie Agent Demo

1. Prerequisites

- Databricks Community Edition account.
- A small dataset uploaded to **Unity Catalog** (e.g., `student_scores.csv` or `retail_sales.csv`).
- A **SQL Warehouse** (Serverless or Pro) — required for Genie Agents.

2. Upload Dataset

- Go to **Data** → **Add Data** → **Upload File**.
- Upload `student_scores.csv` (columns: `student_id`, `name`, `gender`, `math_score`, `reading_score`, `writing_score`).
- Register the dataset in **Unity Catalog** under a schema (e.g., `default.students`).

3. Create Genie Agent

1. In Databricks workspace, go to **Genie Agents** (under AI/BI section).
2. Click **Create Agent**.
3. Configure:
 - **Name:** `Student Performance Agent`
 - **Datasets:** Add `default.students` table.
 - **SQL Warehouse:** Select your running warehouse.
 - **Description:** “Agent to query student performance data.”

4. Ask Natural-Language Questions

Students can now interact with the agent:

Basic Query

Code

[Show all student records.](#)

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-

Aggregation

Code

What is the average math score by gender?

-

Top Results

Code

Show top 5 students by reading score.

-

Conditional Query

Code

List students who scored above 80 in writing.

-

5. View SQL Translation

- Genie will generate SQL automatically, e.g.:

sql

```
SELECT gender, AVG(math_score) AS avg_math
FROM students
GROUP BY gender;
```

Students can compare Genie's SQL with their own manual queries.

6. Visualize Results

- Genie supports charts directly.
- Example: Bar chart of average math score by gender.

7. Learning Outcomes

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- Students experience **natural-language querying** in Databricks.
- They see how Genie auto-generates SQL, bridging technical and non-technical users.
- They understand how conversational BI integrates with **modern lakehouse workflows**.

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Output:

Lab 1: Spark SQL Basics

Spark-SQL-Basics-in-Databricks-01

Delta-Lake-Time-Travel-02

Streaming-Data-Simulation-in-Databricks-03

DataFrame-Transformations-in-Databricks-04

FileEditViewRunHelpPythonFullLast edit was 3 days ago

Run allServerlessScheduleCommentsShare

Python

3 days ago (3s)

1

```
1 # Load CSV into DataFrame
2 df = spark.read.csv("/Workspace/Users/shashikalagupta.rajesh@gmail.com/Drafts/Expanded_data_with_more_features.csv", header=True, inferSchema=True)
3 # Show first 5 rows
4 df.show(5)
```

See performance (1)✓ Checked for improvements

df: pyspark.sql.connect.dataframe.DataFrame = [_c0: integer, Gender: string ... 13 more fields]

_c0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	WklyStudyHours	MathScore	ReadingSc
0	female	NULL	bachelor's degree	standard	none	married	regularly	yes	3	school_bus	< 5	71	
1	female	group C	some college	standard	NULL	married	sometimes	yes	0	NULL	5 - 10	69	
2	female	group B	master's degree	standard	none	single	sometimes	yes	4	school_bus	< 5	87	
3	male	group A	associate's degree	free/reduced	none	married	never	no	1	NULL	5 - 10	45	
4	male	group C	some college	standard	none	married	sometimes	yes	0	school_bus	5 - 10	76	

only showing top 5 rows

3 days ago (<1s)

2

```
1 # Register DataFrame as SQL view
2 df.createOrReplaceTempView("student")
```

See performance (1)✓ Checked for improvements

3 days ago (2s)

3

```
1 %sql
2 -- Select all records
3 SELECT * FROM student;
```

See performance (1)✓ Checked for improvements

Spark-SQL-Basics-in-Databricks-01

Delta-Lake-Time-Travel-02

Streaming-Data-Simulation-in-Databricks-03

DataFrame-Transformations-in-Databricks-04

FileEditViewRunHelpPythonFullLast edit was 3 days ago

Run allServerlessScheduleCommentsShare

Python

3 days ago (2s)

See performance (1)✓ Checked for improvements

Table

+SearchFilterColumns

	_c0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus	PracticeSport	IsFirstChild
1	0	female	NULL	bachelor's degree	standard	none	married	regularly	yes
2	1	female	group C	some college	standard	NULL	married	sometimes	yes
3	2	female	group B	master's degree	standard	none	single	sometimes	yes
4	3	male	group A	associate's degree	free/reduced	none	married	never	no
5	4	male	group C	some college	standard	none	married	sometimes	yes
6	5	female	group B	associate's degree	standard	none	married	regularly	yes
7	6	female	group B	some college	standard	completed	widowed	never	no
8	7	male	group B	some college	free/reduced	none	married	sometimes	yes
9	8	male	group D	high school	free/reduced	completed	single	sometimes	no
10	9	female	group B	high school	free/reduced	none	married	regularly	yes
11	10	male	group C	associate's degree	standard	none	NULL	sometimes	yes
12	11	male	group D	associate's degree	standard	none	divorced	sometimes	yes
13	12	female	group B	high school	standard	none	married	regularly	no
14	13	male	group A	some college	standard	completed	single	sometimes	yes
15									

10,000+ rowsTruncated data2.162s runtime

Refreshed at 10/6/26, 9:04 PM (3d ago)

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Spark-SQL-Basics-in-Databricks-01
Delta-Lake-Time-Travel-02
Streaming-Data-Simulation-in-Databricks-03
DataFrame-Transformations-in-Databricks-04

File Edit View Run Help Python Full Last edit was 3 days ago
Run all Serverless Schedule Comments Share

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

3 days ago (3s) 6

```

1 %sql
2 -- Top 5 students by reading score
3 SELECT _c0, ReadingScore
4 FROM student
5 ORDER BY ReadingScore DESC
6 LIMIT 5;
7

```

See performance (1) Checked for improvements

Table Visualization 1

	<code>_c0</code>	ReadingScore
1	179	100
2	149	100
3	165	100
4	106	100
5	458	100

Download 5 rows 3.068s runtime Refreshed at 10/6/26, 9:10 PM (3d ago)

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

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Lab 2: Delta Lake Time Travel

Lab 02 : Delta Lake Time Travel

Create a Sample Dataset

```

1 # Create sample data
2 data = [
3     (1, "Shashikala", 100),
4     (2, "Gupta", 99),
5     (3, "Alhiks", 98)
6 ]
7 columns = ["student_id", "name", "score"]
8 df = spark.createDataFrame(data, columns)
9 df.show()

```

See performance (1) ✓ Checked for improvements

student_id	name	score
1	Shashikala	100
2	Gupta	99
3	Alhiks	98

Search data, notebooks, recents, and more... CTRL + P

Verify identity workspace

File Edit View Run Help Python Full Last edit was 2 minutes ago

```

1 %sql
2 show catalogs

```

See performance (1) ✓ Checked for improvements

Table	catalog
1	fmcg
2	samples
3	system
4	workspace

4 rows 6.140s runtime Refreshed at 10/9/26, 9:44 PM (4m ago)

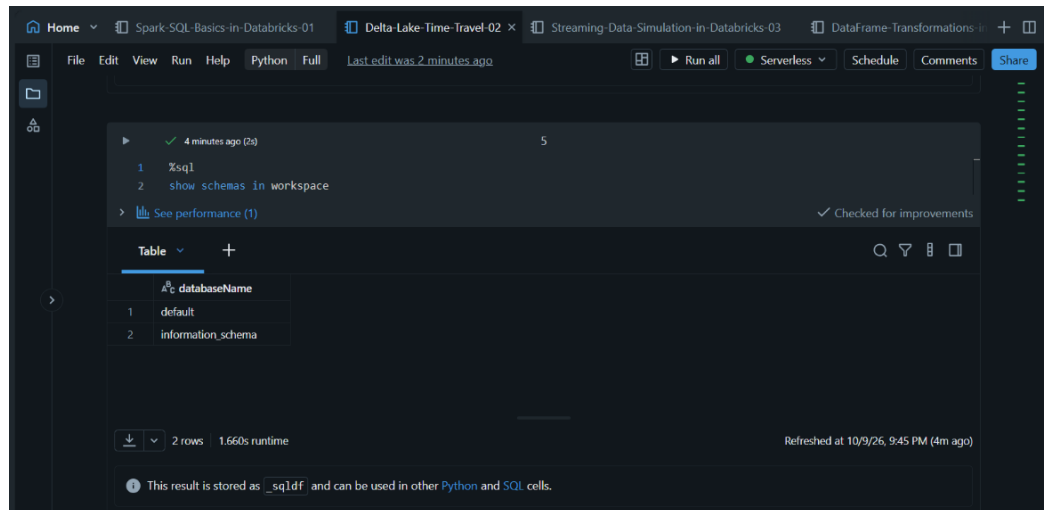
This result is stored as `_sqldf` and can be used in other Python and SQL cells.

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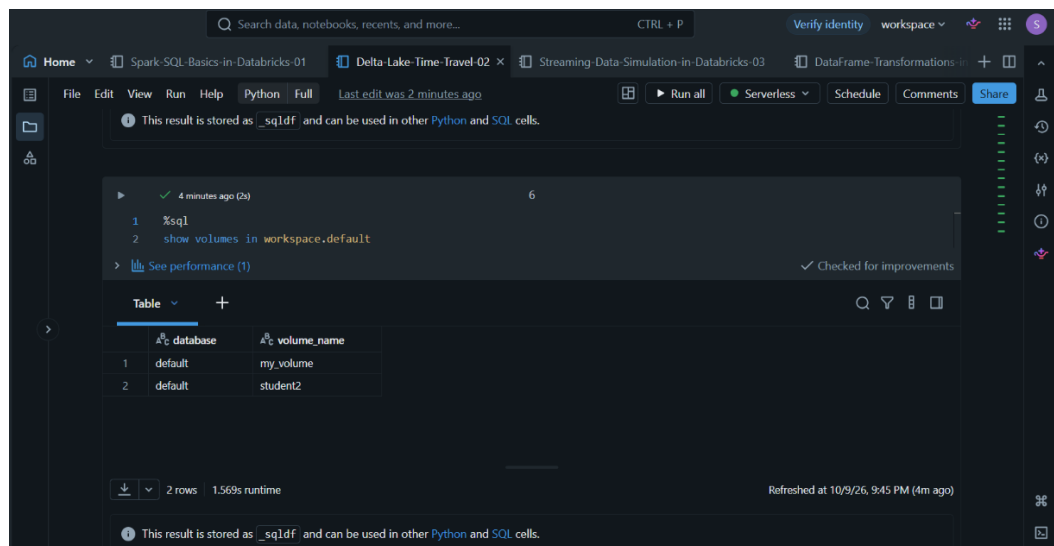
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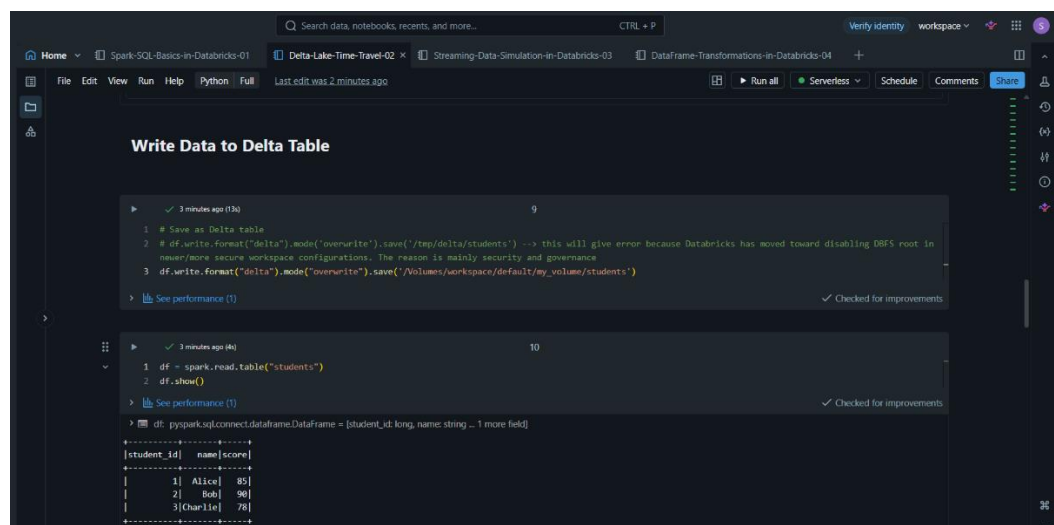
The screenshot shows a Databricks notebook titled "Delta Lake Time-Travel-02". The code cell contains a SQL query: `%sql` followed by `show schemas in workspace`. The output is a table with two rows: `default` and `information_schema`. The table has a header `databaseName`. The runtime is 1.660s. The result is stored as `_sql_df`.

databaseName
default
information_schema



The screenshot shows a Databricks notebook titled "Delta Lake Time-Travel-02". The code cell contains a SQL query: `%sql` followed by `show volumes in workspace, default`. The output is a table with two rows: `default` and `default`. The table has headers `database` and `volume_name`. The runtime is 1.569s. The result is stored as `_sql_df`.

database	volume_name
default	my_volume
default	student2



The screenshot shows a Databricks notebook titled "Delta Lake Time-Travel-02". The code cell contains Python code to write data to a Delta table. The code is as follows:

```
1 # Save as Delta table
2 # df.write.format("delta").mode("overwrite").save("/tmp/delta/students") --> this will give error because Databricks has moved toward disabling DBFS root in
   newer/more secure workspace configurations. The reason is mainly security and governance
3 df.write.format("delta").mode("overwrite").save("/Volumes/workspace/default/my_volume/students")
```

The output is a table with three rows: `1| Alice| 85`, `2| Bob| 90`, and `3| Charlie| 78`. The table has headers `student_id`, `name`, and `score`. The runtime is 3 minutes ago (13s).

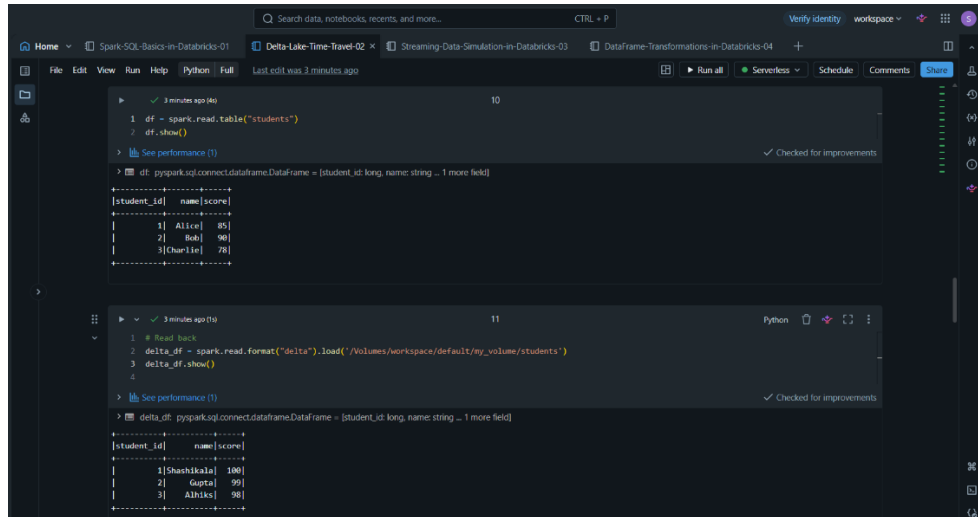
student_id	name	score
1	Alice	85
2	Bob	90
3	Charlie	78

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```
1 df = spark.read.table("students")
2 df.show()

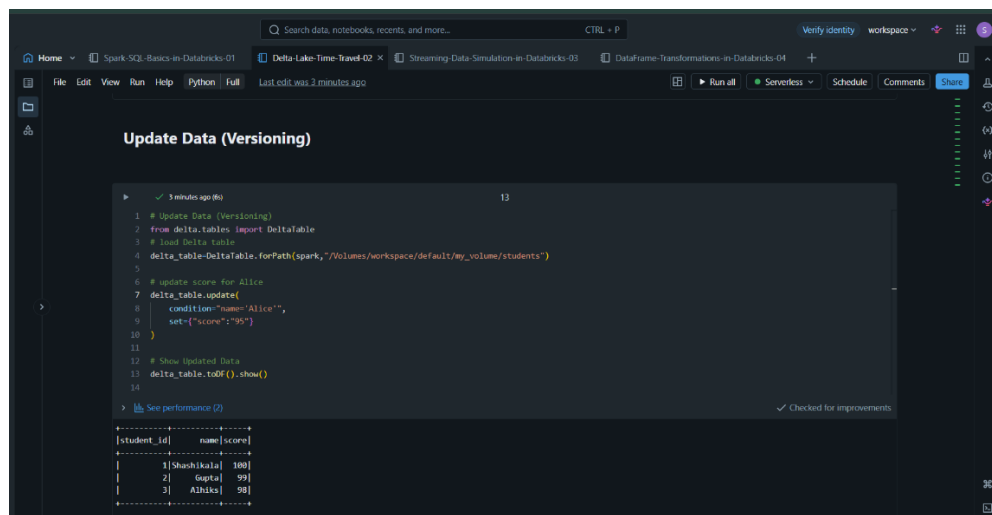
> dt: pyspark.sql.connect.dataframe.DataFrame = [student_id: long, name: string ... 1 more field]

[student_id] [name] [score]
-----
[1] [Alice] [85]
[2] [Bob] [90]
[3] [Charlie] [78]
```

```
1 # Read back
2 delta_df = spark.read.format("delta").load("/Volumes/workspace/default/my_volume/students")
3 delta_df.show()

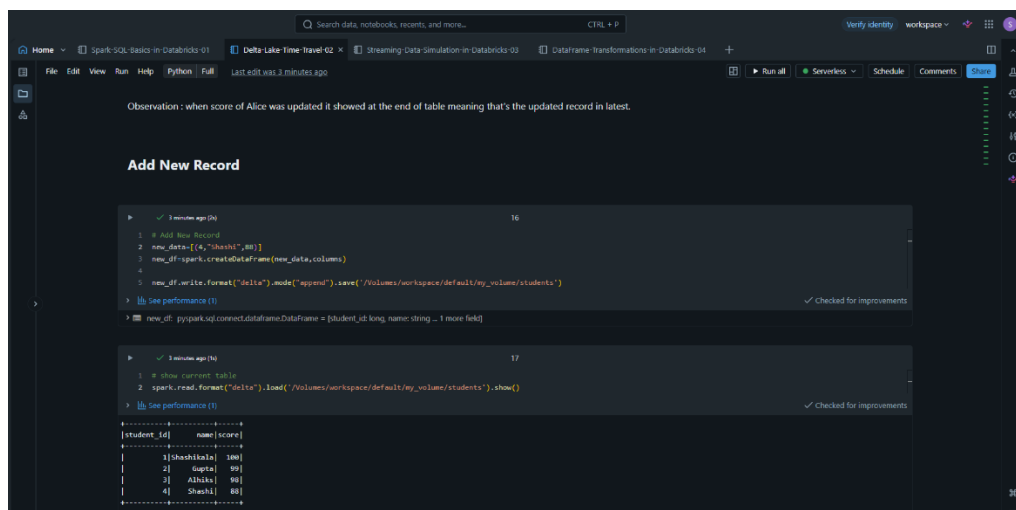
> delta_df: pyspark.sql.connect.dataframe.DataFrame = [student_id: long, name: string ... 1 more field]

[student_id] [name] [score]
-----
[1] [Shashikala] [100]
[2] [Gupta] [99]
[3] [Ahhiks] [98]
```



```
1 # Update Data (Versioning)
2 from delta.tables import DeltaTable
3 # Load Delta Table
4 delta_table = DeltaTable.forPath(spark, "/Volumes/workspace/default/my_volume/students")
5
6 # update score for Alice
7 delta_table.update(
8     condition="name='Alice'",
9     set={"score": 99}
10 )
11
12 # Show Updated Data
13 delta_table.toDF().show()

[student_id] [name] [score]
-----
[1] [Shashikala] [100]
[2] [Gupta] [99]
[3] [Ahhiks] [98]
```



Observation: when score of Alice was updated it showed at the end of table meaning that's the updated record in latest.

```
1 # Add New Record
2 new_data = [{"id": "Shashi", "ss": 88}]
3 new_df = spark.createDataFrame(new_data, columns)
4
5 new_df.write.format("delta").mode("append").save("/Volumes/workspace/default/my_volume/students")

> new_df: pyspark.sql.connect.dataframe.DataFrame = [student_id: long, name: string ... 1 more field]
```

```
1 # show current table
2 spark.read.format("delta").load("/Volumes/workspace/default/my_volume/students").show()

[student_id] [name] [score]
-----
[1] [Shashikala] [100]
[2] [Gupta] [99]
[3] [Ahhiks] [98]
[4] [Shashi] [88]
```

Name: Shashikala Rajesh Gupta

Roll Number: 12

Year/Dept:
BE COMP 2026-27

Subject:
Big Data Analytics (Assignment)

The screenshot shows a Databricks workspace with a notebook titled "Query older versions". The notebook contains two code blocks. The first code block, executed 2 minutes ago, shows the command `delta.table.history().show()` and its output, which is a table with columns: version, timestamp, userId, userTime, operation, operationParameters, job, notebook, queryHistoryStatementId, clusterId, readVersion, and isLocal. The second code block, also executed 2 minutes ago, shows the command `spark.read.format("delta").option("versionAsOf", 0).load("/Volumes/workspace/default/my_volume/students").show()` and its output, which is a table with columns: student_id, name, and score.

```
1 # Check history
2 delta.table.history().show()
```

version	timestamp	userId	userTime	operation	operationParameters	job	notebook	queryHistoryStatementId	clusterId	readVersion	isLocal
6	2026-10-09 16:16:59	74162278981546	shashikalgupta.r...	WRITE	{mode -> Append, ...}	MULTI	[3916570572513088]	2c1fdd67-0420-473...	1000-163424-3667...	0	WriteSerial
5	2026-10-09 16:16:35	74162278981546	shashikalgupta.r...	UPDATE	{predicate -> ["(...	MULTI	[3916570572513088]	5b7ed1d6-0200-4b5...	1000-163424-3667...	4	WriteSerial
4	2026-10-09 16:16:11	74162278981546	shashikalgupta.r...	WRITE	{mode -> Overwrite...	MULTI	[3916570572513088]	a74c6acc-7332-4bf...	1000-163424-3667...	3	WriteSerial
3	2026-10-09 17:21:06	74162278981546	shashikalgupta.r...	WRITE	{mode -> Append, ...}	MULTI	[3916570572513088]	d448d853-5a26-4fd...	1000-152440-qcufp...	2	WriteSerial
2	2026-10-06 17:02:11	74162278981546	shashikalgupta.r...	OPTIMIZE	{clusterBy -> [], ...}	MULTI	[3916570572513088]	8009b216-ca82-417...	1000-152440-qcufp...	1	SnapshotIs
1	2026-10-06 17:02:28	74162278981546	shashikalgupta.r...	UPDATE	{predicate -> ["(...	MULTI	[3916570572513088]	8009b216-ca82-417...	1000-152440-qcufp...	0	WriteSerial
0	2026-10-06 16:51:21	74162278981546	shashikalgupta.r...	WRITE	{mode -> Overwrite...	MULTI	[3916570572513088]	7e68d4de-5642-4eb...	1000-152440-qcufp...	NULL	WriteSerial

```
1 # Read Version 0 (original data)
2 spark.read.format("delta").option("versionAsOf", 0).load("/Volumes/workspace/default/my_volume/students").show()
```

student_id	name	score
1	Alice	85
2	Bob	90
3	Charlie	78

The screenshot shows a Databricks workspace with a notebook titled "Delta Lake-Time-Travel-02". The notebook contains two code blocks. The first code block, executed 3 minutes ago, shows the command `spark.read.format("delta").option("versionAsOf", 0).load("/Volumes/workspace/default/my_volume/students").show()` and its output, which is a table with columns: student_id, name, and score. The second code block, executed 2 minutes ago, shows the command `spark.read.format("delta").option("versionAsOf", 1).load("/Volumes/workspace/default/my_volume/students").show()` and its output, which is a table with columns: student_id, name, and score.

```
1 # Read Version 0 (original data)
2 spark.read.format("delta").option("versionAsOf", 0).load("/Volumes/workspace/default/my_volume/students").show()
```

student_id	name	score
1	Alice	85
2	Bob	90
3	Charlie	78

```
1 # Read Version 1 (after alice update)
2 spark.read.format("delta").option("versionAsOf", 1).load("/Volumes/workspace/default/my_volume/students").show()
```

student_id	name	score
2	Bob	90
3	Charlie	78
1	Alice	95

Name: Shashikala Rajesh Gupta	Roll Number: 12
Year/Dept: BE COMP 2026-27	Subject: Big Data Analytics (Assignment)

Lab 3: Streaming Data Simulation

Search data, notebooks, recents, and more... CTRL + P

Verify identity workspace

Home Spark-SQL-Basics-in-Databricks-01 Delta-Lake-Time-Travel-02 **Streaming-Data-Simulation-in-Databricks-03** DataFrame-Transformations-in-Databricks-04

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Lab 03 : Streaming Data Simulation in Databricks

Create a simulated streaming source

3 days ago (<1s) 3

```
1 import time
2 import json
```

create a directory for streaming data

3 days ago (1s) 5

```
1 dbutils.fs.mkdirs('/Volumes/workspace/default/my_volume/streaming_data/')
> See performance (1) ✓ Checked for improvements
```

True

Search data, notebooks, recents, and more... CTRL + P

Verify identity workspace

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Simulate writing JSON events

3 days ago (3s) 7

```
1 for i in range(5):
2     event={"event_id":i,"sensor_value":i*10,"timestamp":str(time.time())}
3     path=f'/Volumes/workspace/default/my_volume/streaming_data/event_{i}.json'
4     dbutils.fs.put(path,json.dumps(event),overwrite=True)
> See performance (5) ✓ Checked for improvements
```

Wrote 69 bytes.
Wrote 70 bytes.
Wrote 70 bytes.
Wrote 70 bytes.
Wrote 70 bytes.

Read Stream with Structured Streaming

3 days ago (1s) 9

```
1 stream_df=(
2     spark.readStream.format("json").schema("event_id INT,sensor_value INT,timestamp STRING").load('/Volumes/workspace/default/my_volume/streaming_data')
3 )
> stream_df: pyspark.sql.connect.dataframe.DataFrame = [event_id: integer, sensor_value: integer ... 1 more field]
```

Name: Shashikala Rajesh Gupta	Roll Number: 12
Year/Dept: BE COMP 2026-27	Subject: Big Data Analytics (Assignment)

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Display stream in real-time

3 days ago 11

```
1 display(stream_df,checkpointLocation="/Volumes/workspace/default/my_volume/streaming_data/checkpoint")
```

> [See performance \(3\)](#) ✓ Checked for improvements

> display_query_3 (id: 1f664b7c-0d02-4021-85c8-85221295c5d) Last updated: 3 days ago

Checkpointing to /Volumes/workspace/default/my_volume/streaming_data/checkpoint

Process Streaming Data

3 days ago (11)

```
1 # Aggregate sensor values
2 agg_df=(
3     stream_df.groupBy("event_id").agg(("sensor_value": "avg")),withColumnRenamed("avg(sensor_value)","avg_value")
4 )
```

> agg_df: pyspark.sql.connect.dataframe.DataFrame = [event_id: integer, avg_value: double]

3 days ago 14

```
1 # Write results to memory table
2 query=(
3     agg_df.writeStream.outputMode("complete").format("memory").option("checkpointLocation", "/Volumes/workspace/default/my_volume/checkpoint").trigger(availableNow=True).queryName("sensor_avg").start()
4 )
```

> [See performance \(1\)](#) Optimize

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Query Results

3 days ago (15) 16

```
1 %sql
2 -- Run this in a SQL cell
3 select * from sensor_avg;
```

> [See performance \(1\)](#) ✓ Checked for improvements

Table Visualization 1

	1: event_id	1:2 avg_value
1	1	10
2	3	30
3	4	40
4	2	20
5	0	0

5 rows 1.022s runtime Refreshed at 10/7/26, 12:15 AM (3d ago)

This result is stored as _sqldf and can be used in other Python and SQL cells.

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Lab 4: DataFrame Transformations

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Lab 04: DataFrame Transformations in Databricks

Load Sample Dataset

```

1 df = spark.read.csv('/Workspace/Users/shashikalagupta.rajesh@gmail.com/Drafts/Expanded_data_with_more_features.csv', header=True,
2   inferSchema=True)
3 df.show(5)

```

See performance (1)

df: pyspark.sql.connect.dataframe.DataFrame = [c0: integer, Gender: string ... 13 more fields]

_c0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	WklyStudyHrs
0	female	NULL	bachelor's degree	standard	none	married	regularly	yes	3	school_bus	5
1	female	group C	some college	standard	NULL	married	sometimes	yes	0	NULL	5
2	female	group B	master's degree	standard	none	single	sometimes	yes	4	school_bus	5
3	male	group A	associate's degree	free/reduced	none	married	never	no	1	NULL	5
4	male	group C	some college	standard	none	married	sometimes	yes	0	school_bus	5

only showing top 5 rows

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Filter Rows

```

1 # Filter students with math score >80
2 high_math = df.filter(df.MathScore > 80)
3 high_math.show(5)

```

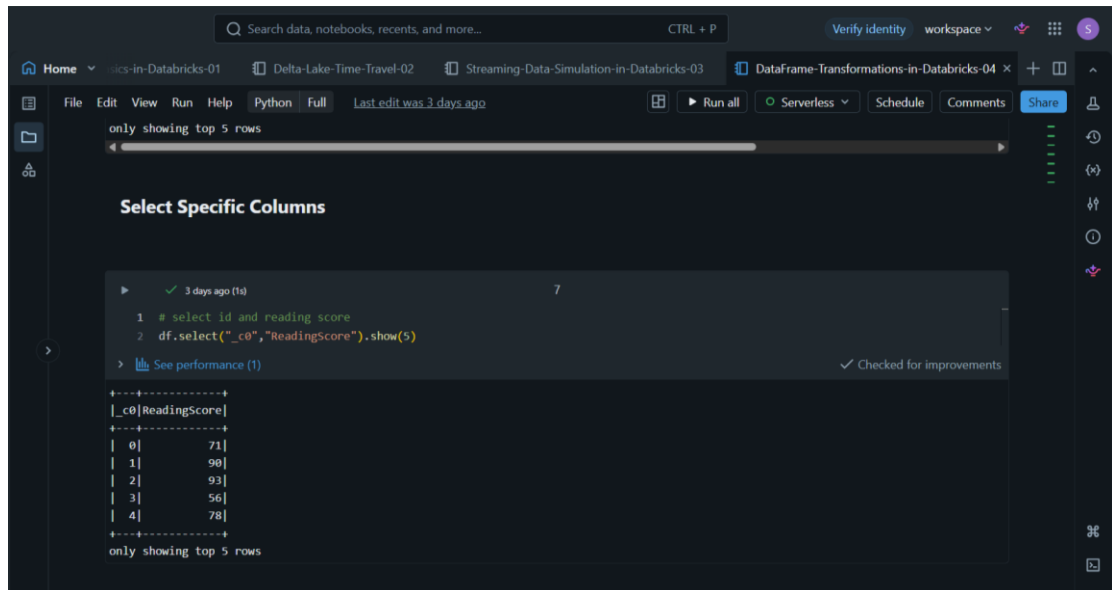
See performance (1)

high_math: pyspark.sql.connect.dataframe.DataFrame = [c0: integer, Gender: string ... 13 more fields]

_c0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	WklyStudyHrs
2	female	group B	master's degree	standard	none	single	sometimes	yes	4	school_bus	5
6	female	group B	some college	standard	completed	widowed	never	no	1	private	5
16	male	group C	high school	standard	NULL	married	sometimes	yes	0	school_bus	5
34	male	group E	some college	standard	none	divorced	sometimes	yes	0	school_bus	5
35	male	group E	associate's degree	standard	completed	single	sometimes	no	NULL	school_bus	5

only showing top 5 rows

Name: Shashikala Rajesh Gupta	Roll Number: 12
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only showing top 5 rows

Select Specific Columns

```

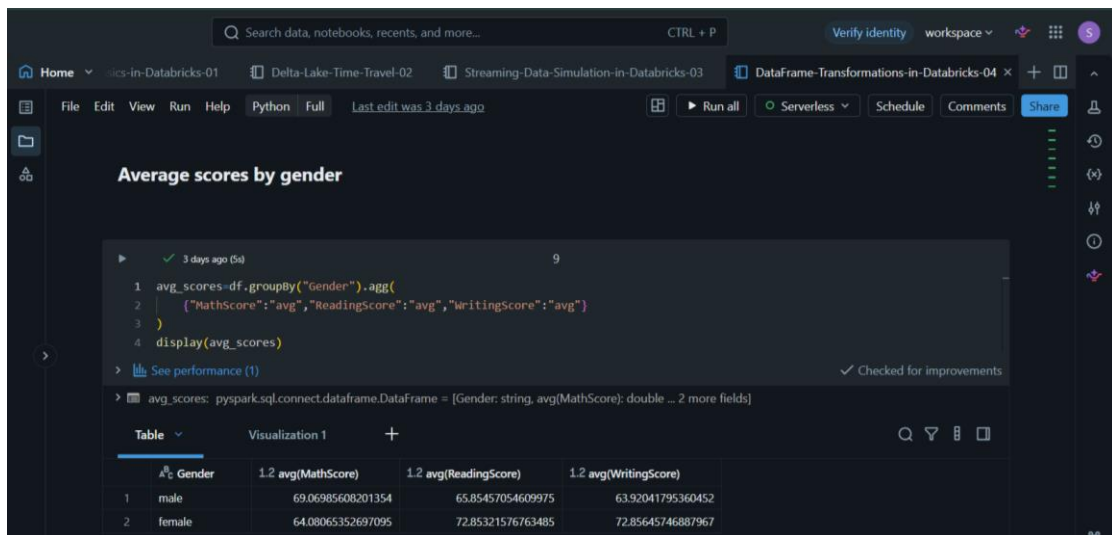
1 # select id and reading score
2 df.select("_c0","ReadingScore").show(5)

```

See performance (1) ✓ Checked for improvements

_c0	ReadingScore
0	71
1	90
2	93
3	56
4	78

only showing top 5 rows



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Average scores by gender

```

1 avg_scores=df.groupby("Gender").agg(
2   ("MathScore":"avg","ReadingScore":"avg","WritingScore":"avg")
3 )
4 display(avg_scores)

```

See performance (1) ✓ Checked for improvements

avg_scores: pyspark.sql.connect.dataframe.DataFrame = [Gender: string, avg(MathScore): double ... 2 more fields]

Gender	1.2 avg(MathScore)	1.2 avg(ReadingScore)	1.2 avg(WritingScore)
male	69.06985608201354	65.85457054609975	63.92041795360452
female	64.08065352697095	72.85321576763485	72.85645746887967

Name: Shashikala Rajesh Gupta

Roll Number: 12

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```
Join Two DataFrames

1 # Create another DataFrame with student hobbies
2 hobbies = [
3     (1, "Football"),
4     (2, "Chess"),
5     (3, "Music"),
6     (4, "Dance")
7 ]
8
9 hobby_df = spark.createDataFrame(hobbies, ['_c0', 'hobby'])
10
11 hobby_df: pyspark.sql.connect.dataframe.DataFrame = [_c0: long, hobby: string]

1 # Join with main DataFrame
2 joined_df = df.join(hobby_df, "_c0", "left")
3 joined_df.show()
```

```
joined_df: pyspark.sql.connect.dataframe.DataFrame = [_c0: integer, Gender: string ... 14 more fields]
```

_c0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus	PracticeSport	IsFirstChild	nrSiblings	TransportMode
0	female	NULL	bachelor's degree	standard	none	married	regularly	yes	3	school
1	female	group C	some college	standard	NULL	married	sometimes	yes	0	
2	female	group B	master's degree	standard	none	single	sometimes	yes	4	school
3	male	group A	associate's degree	free/reduced	none	married	never	no	1	
4	male	group C	some college	standard	none	married	sometimes	yes	0	school
5	female	group B	associate's degree	standard	none	married	regularly	yes	1	school
6	female	group B	some college	standard/completed	none	widowed	never	no	1	pri
7	male	group B	some college	free/reduced	none	married	sometimes	yes	1	pri
8	male	group D	high school	free/reduced	completed	single	sometimes	no	3	pri
9	female	group D	high school	free/reduced	none	married	regularly	yes	NULL	pri
10	male	group C	associate's degree	standard	none	NULL	sometimes	yes	1	pri
11	male	group B	associate's degree	standard	none	divorced	sometimes	yes	1	school
12	female	group D	high school	standard	none	married	regularly	no	1	pri
13	male	group A	some college	standard/completed	single	sometimes	yes	1	1	pri
14	female	group A	master's degree	standard	none	divorced	sometimes	yes	2	pri
15	female	group C	some high school	standard	none	married	sometimes	yes	0	pri
16	male	group C	high school	standard	NULL	married	sometimes	yes	0	school

```
1 # Top 5 students writing score
2 df.orderBy(df.writingScore.desc()).show(5)
```

_c0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus	PracticeSport	IsFirstChild	nrSiblings	TransportMode
106	female	group D	master's degree	standard	none	married	sometimes	no	3	privat
377	female	group D	master's degree	free/reduced/completed	single	sometimes	yes	NULL	0	privat
165	female	group C	bachelor's degree	standard/completed	married	sometimes	yes	0	0	school_bu
114	female	group E	bachelor's degree	standard/completed	married	regularly	yes	3	3	school_bu
179	female	group D	some high school	standard/completed	divorced	regularly	yes	0	0	school_bu

only showing top 5 rows

Name: Shashikala Rajesh Gupta	Roll Number: 12
Year/Dept: BE COMP 2026-27	Subject: Big Data Analytics (Assignment)

Lab 5: Genie Agent Demo

Expanded_data_with_more_features.csv uploaded 2.91MB

Create new table

Preview mode

Catalogworkspace

Schemadefault

Table nameexpanded_data_with_more_features

Advanced attributes

Previewing 50 rows, 14 columns

Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	
female	null	bachelor's degree	standard	none	married
female	group C	some college	standard	null	married
female	group B	master's degree	standard	none	single
male	group A	associate's degree	free/reduced	none	married
male	group C	some college	standard	none	married
female	group B	associate's degree	standard	none	married
female	group B	some college	standard	completed	widowed
male	group B	some college	free/reduced	none	married
male	group D	high school	free/reduced	completed	single
female	group B	high school	free/reduced	none	married
male	group C	associate's degree	standard	none	null

Cancel

Create table

Search data, notebooks, recent, and more...CTRL + P

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ChatMonitorBenchmark

ConfigureShare

Student Performance and Demographics

Ask your question...

AgentChat

Always review the accuracy of responses.

What is the distribution of values in the ParentEduc column?

What tables are there and how are they connected? Give me a short summary.

What is the distribution (min, max, average, median) of MathScore?

What is the distribution of values in the Gender column?

Name: Shashikala Rajesh Gupta

Roll Number: 12

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What tables are there and how are they connected? G Reviewable

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Show all student records.

Thought process

All Student Records

The table below shows all 30,641 student records from the database. You can scroll through the data, sort by any column, and download the full dataset.

Show code

#	Gender	Ethnic Group	Parent Education	Lunch Type	Test Prep	Parent Marital Status	Practice Sport	First Child	# Siblings	Transport
1	female	null	bachelor's degree	standard	none	married	regularly	yes	3	school_bus
2	female	group C	some college	standard	null	married	sometimes	yes	0	null
3	female	group B	master's degree	standard	none	single	sometimes	yes	4	school_bus
4	male	group A	associate's degree	free/reduced	none	married	never	no	1	null
5	male	group C	some college	standard	none	married	sometimes	yes	0	school_bus

Ask your question...

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Show code

#	Gender	Ethnic Group	Parent Education	Lunch Type	Test Prep	Parent Marital Status	Practice Sport	First Child	# Siblings	Transport
1	female	null	bachelor's degree	standard	none	married	regularly	yes	3	school_bus
2	female	group C	some college	standard	null	married	sometimes	yes	0	null
3	female	group B	master's degree	standard	none	single	sometimes	yes	4	school_bus
4	male	group A	associate's degree	free/reduced	none	married	never	no	1	null
5	male	group C	some college	standard	none	married	sometimes	yes	0	school_bus
6	female	group B	associate's degree	standard	none	married	regularly	yes	1	school_bus

< 1 2 3 4 5 ... 100 >

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#	Gender	Ethnic Group	Parent Education	Lunch Type	Test Prep	Parent Marital Status	Practice Sport	First Child	# Siblings	Transport
1	female	null	bachelor's degree	standard	none	married	regularly	yes	3	school_bus
2	female	group C	some college	standard	null	married	sometimes	yes	0	null
3	female	group B	master's degree	standard	none	single	sometimes	yes	4	school_bus
4	male	group A	associate's degree	free/reduced	none	married	never	no	1	null
5	male	group C	some college	standard	none	married	sometimes	yes	0	school_bus
6	female	group B	associate's degree	standard	none	married	regularly	yes	1	school_bus

< 1 2 3 4 5 ... 100 >

Source: All student records

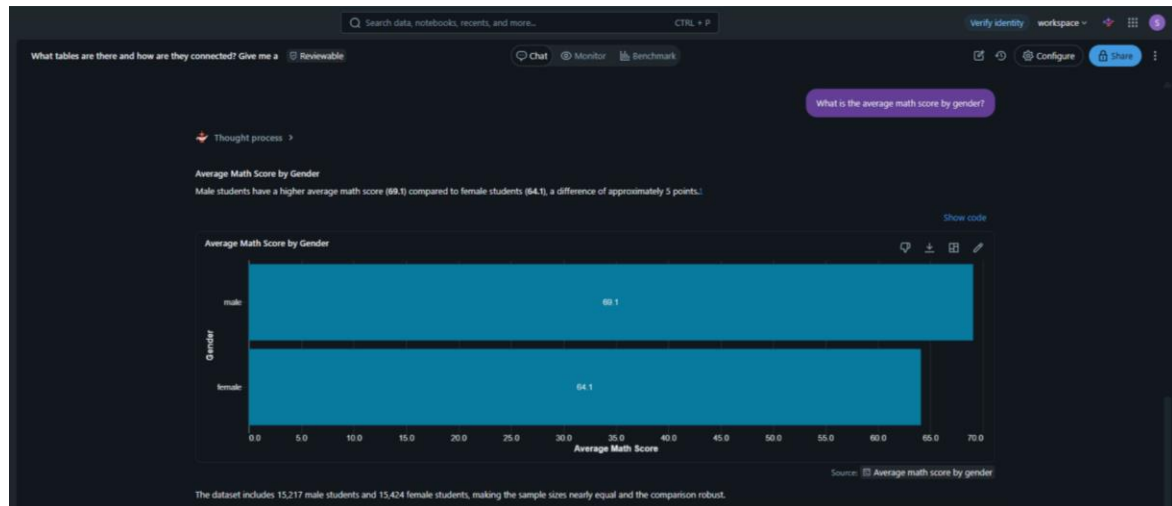
Each record includes demographic information (gender, ethnic group), family background (parent education, marital status, siblings), lifestyle factors (lunch type, test prep, sports, transport, study hours), and academic performance scores (math, reading, writing).

Name: Shashikala Rajesh Gupta

Roll Number: 12

Year/Dept:
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Subject:
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What tables are there and how are they connected? Give me a [Reviewable](#)

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Show top 5 students by reading score.

Thought process

Top 5 Students by Reading Score

All five top-performing students achieved a perfect reading score of 100.

Show code

#	Gender	Ethnic Group	Parent Education	Test Prep	Practice Sport	Weekly Study Hours	Reading Score	Math Score	Writing Score
1	female	group C	bachelor's degree	completed	sometimes	5 - 10	100	96	100
2	male	group E	associate's degree	completed	regularly	5 - 10	100	100	95
3	female	group D	master's degree	none	sometimes	5 - 10	100	87	100
4	female	group D	some high school	completed	regularly	5 - 10	100	99	100
5	female	group E	bachelor's degree	none	regularly	not	100	100	100

Source: Top 5 students by reading score

Key observations

- 4 out of 5 are female students
- 3 out of 5 completed test preparation
- All five also scored exceptionally well in writing (95-100) and math (87-100)
- They come from diverse ethnic groups (C, D, and E) and parent education levels (ranging from "some high school" to "master's degree")

Sources (1)

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List students who scored above 80 in writing.

Thought process

Students with Writing Score Above 80

There are 6,538 students who scored above 80 in writing, representing approximately 23% of all students.

Show code

#	Gender	Ethnic Group	Parent Education	Test Prep	Study Hours	Writing Score	Reading Score	Math Score
1	female	group D	master's degree	completed	5 - 10	100	96	81
2	male	group E	associate's degree	completed	> 10	100	96	100
3	male	group E	master's degree	completed	> 10	100	100	100
4	female	group E	bachelor's degree	completed	5 - 10	100	100	91
5	female	group E	not	completed	5 - 10	100	100	93
6	female	group C	bachelor's degree	completed	5 - 10	100	97	92
7	female	group D	associate's degree	completed	5 - 10	100	100	88

Source: Students with writing score above 80

The table shows all students meeting this criteria, sorted by writing score (highest first). You can scroll through all records, sort by any column, and download the complete list. Scores of 95 or higher are highlighted in green and bold.

Sources (1)

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Sources (1)

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SELECT gender, AVG(math_score) AS avg_math FROM students GROUP BY gender;

Thought process

Query Results: Average Math Score by Gender

I've executed your query (adjusted for the correct table and column names in this database):

1

Results:

Female students: Average math score of 64.1

Male students: Average math score of 69.1

Male students have an approximately 5-point higher average in math compared to female students.

Note: The actual table name is workspace.default.expanded_data_with_more_features and the columns are Gender and MathScore (capitalized), not students, gender, and math_score.

Sources (1)

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What tables are there and how are they connected? Give me a Reviewable

ChatMonitorBenchmark

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pie chart of average math score by gender.

Thought process

Average Math Score by Gender

Average Math Score by Gender

Average Math Score

48.13%

51.87%

Gender

female

male

Show code

Source: Average math score by gender for pie chart

The pie chart shows the distribution of average math scores by gender. Male students have a higher average math score (69.1) compared to female students (64.1), representing approximately 52% vs 48% of the combined average scores.

Sources (1)

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Year/Dept: BE COMP 2026-27	Subject: Big Data Analytics (Assignment)

Conclusion:

Lab 1: Spark SQL Basics in Databricks

I learned how to upload CSV datasets into Databricks, load them into Spark DataFrames, create temporary SQL views, and execute SQL queries using SELECT, GROUP BY, ORDER BY, and aggregate functions. I also learned how to visualize query results using charts.

Lab 2: Delta Lake Time Travel

I learned how to create and manipulate Delta tables, update existing records, append new data, and view previous versions of a dataset using time travel. I understood how Delta Lake supports ACID transactions and maintains data version history for reliable data management.

Lab 3: Delta Lake Time Travel

I learned how to simulate streaming data using JSON files and process it with Spark Structured Streaming. I understood how to read incoming events, perform aggregations on sensor data, and display the results through a memory table. This helped me understand the basics of real-time data processing.

Lab 4: DataFrame Transformations

I learned how to perform data manipulation using PySpark DataFrames, including filtering rows, selecting columns, grouping data, applying aggregate functions, joining multiple DataFrames, and sorting records. I also learned how to visualize the processed data to identify useful patterns and insights.

Lab 5: Genie Agent Demo

I learned how to use Databricks Genie to query datasets using natural-language questions. I understood how conversational analytics can generate SQL queries automatically, retrieve relevant results, and create visualizations. This introduced me to AI-assisted business intelligence and made data analysis more accessible to non-technical users.